

PROJECT: BEE-1012

sub-system: EXP

Version: 1.0.0



Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH		Sheet title: 01_COVER PAGE.SchDocSheet 1 of 25	
Checked by: BAO BUI Q.		Document No: 1	Size: A3
Approved by: VU PHAM		Revision: V1.2.2	Date: 12/2/2025

1	2	3	4	5	6
A					
B					
C					
D					
1	2	3	4	5	6



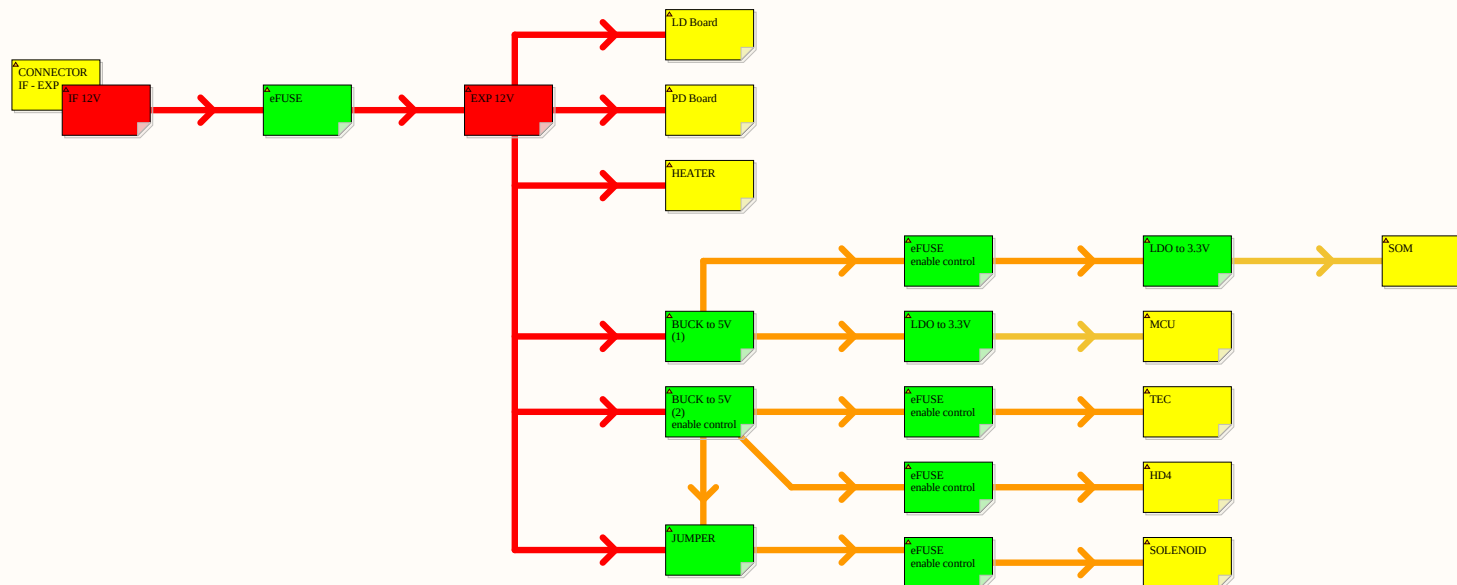
Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 02_BLOCK DIAGRAM.Sch	Sheet 2 of 25
Checked by: BAO BUI Q.	Document No: 2	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

INDEX

Page	Title	Page	Title	Page	Title
1	COVER PAGE	11		21	
2	BLOCK DIAGRAM	12		22	
3	INDEX	13		23	
4	PI HAT	14		24	
5		15		25	
6		16		26	
7		17		27	
8		18		28	
9		19		29	
10		20		30	

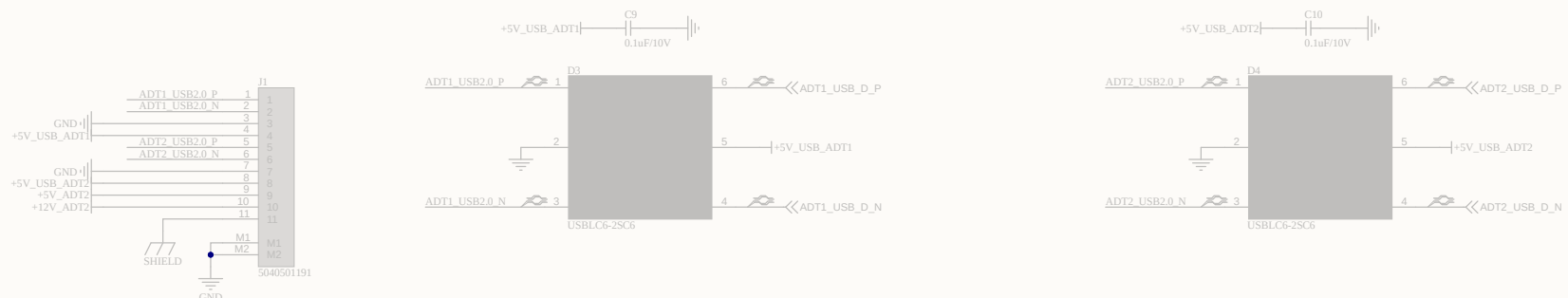


Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 03_INDEX.SchDoc	Sheet 3 of 25	
Checked by: BAO BUI Q.	Document No: 3	Size: A3	
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025	

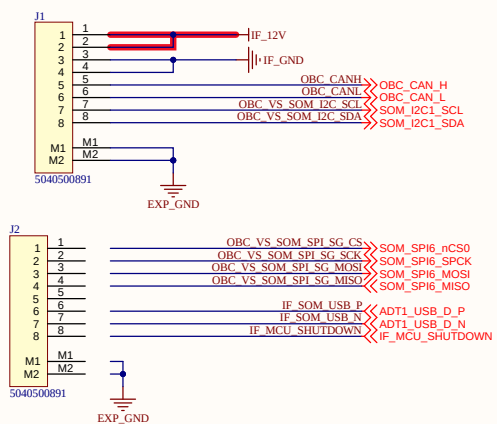


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 04_POWER_TREE.SchDoc	Sheet 4 of 25
Checked by: BAO BUI Q.	Document No: 4	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

form Adaptor Board

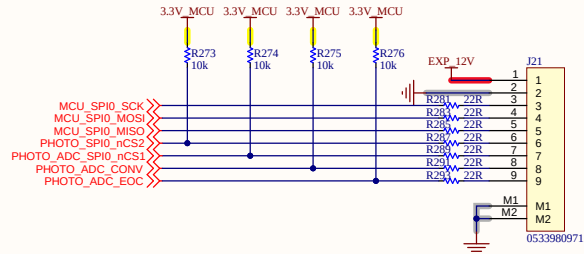


form IF (OBC Board)

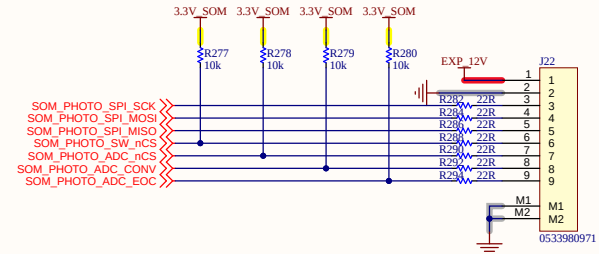


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 05_CONNECTOR_IN.SchDoc	Sheet 5 of 25
Checked by: BAO BUI Q.	Document No: 5	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

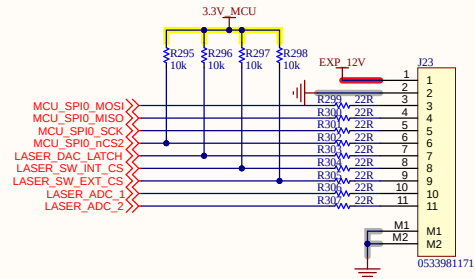
for PHOTO board



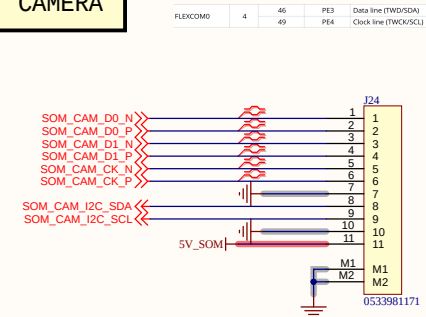
SOM to PHOTO



for LASER board



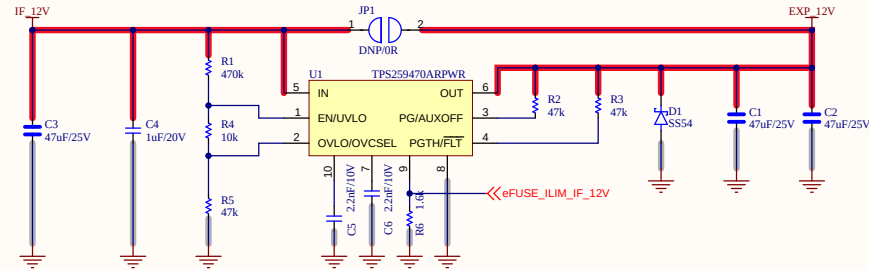
for CAMERA



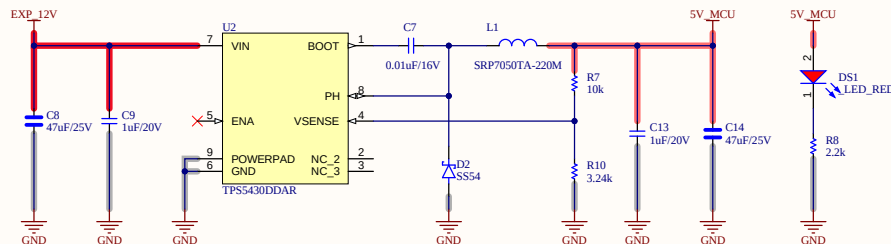
Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 05_CONNECTOR_OUT.Schema	Sheet 6 of 25
Checked by: BAO BUI Q.	Document No: 6	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

EFUSE 12V

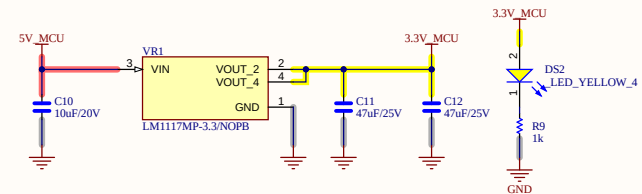
Undervoltage Protector: NONE
 Overvoltage Protector: $1.2 \times (47 + 470) / 47 = 13.2V$
 Slew Rate: $2000 / 2200 = 0.91V/ms$
 I-LIM: $3334 / 1600 = 2.08A$



BUCK for MCU, SOM

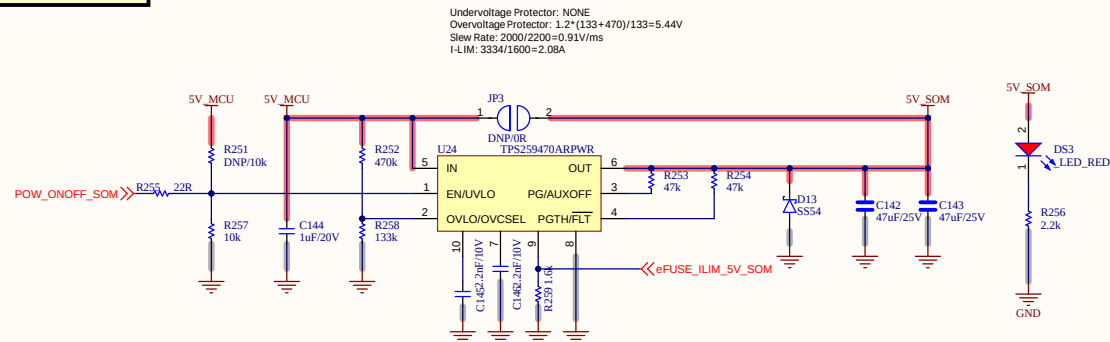


LDO for MCU

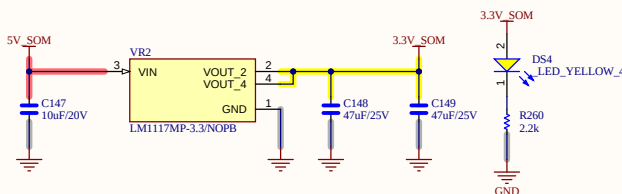


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 06_POWER_1.SchDoc	Sheet 7 of 25
Checked by: BAO BUI Q.	Document No: 7	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

EFUSE for SOM

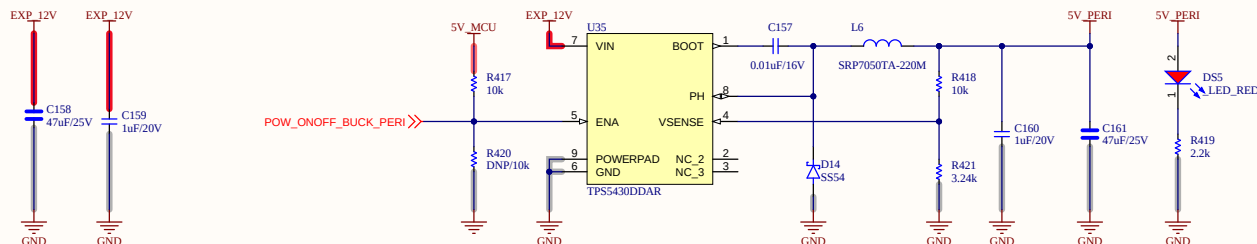


LD0 for SOM

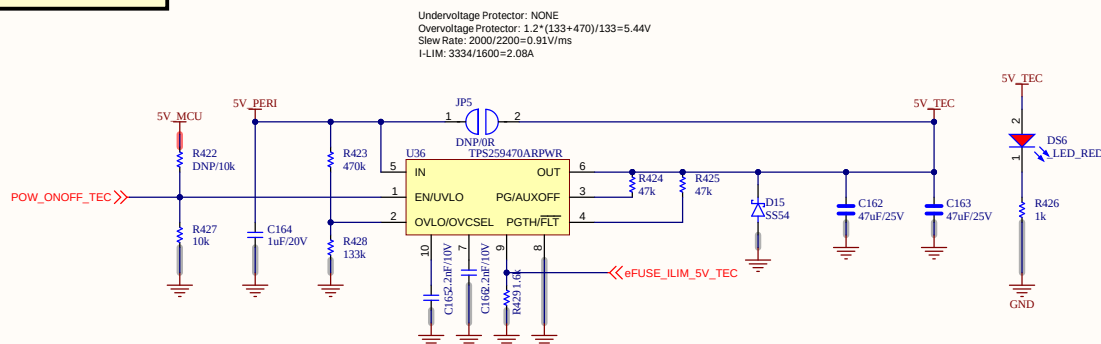


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 06_POWER_2.SchDoc	Sheet 8 of 25
Checked by: BAO BUI Q.	Document No: 8	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

BUCK for PERIPHERAL

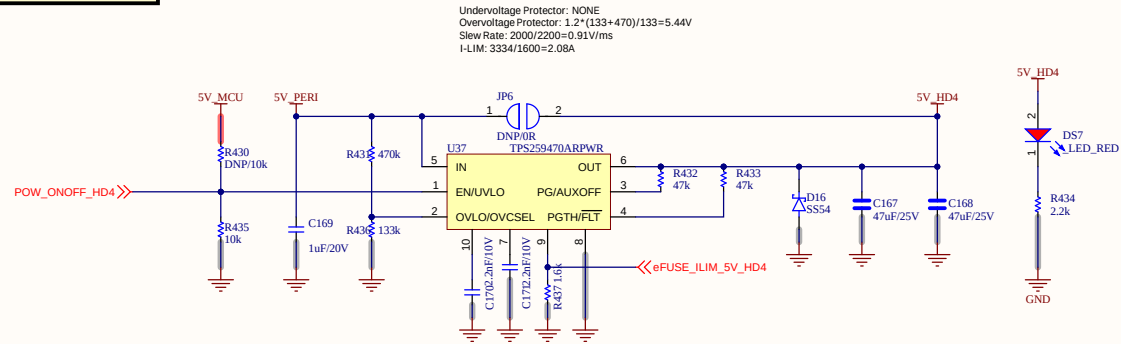


EFUSE for TEC

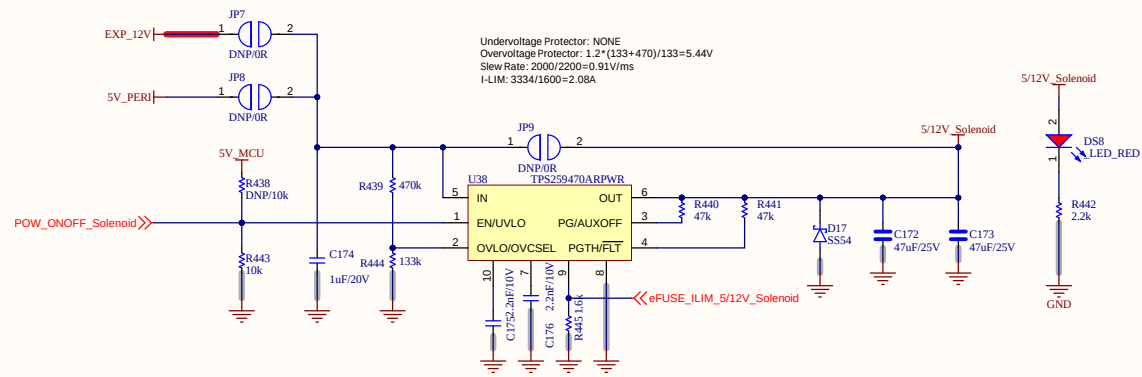


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 06_POWER_3.SchDoc	Sheet 9 of 25
Checked by: BAO BUI Q.	Document No: 9	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

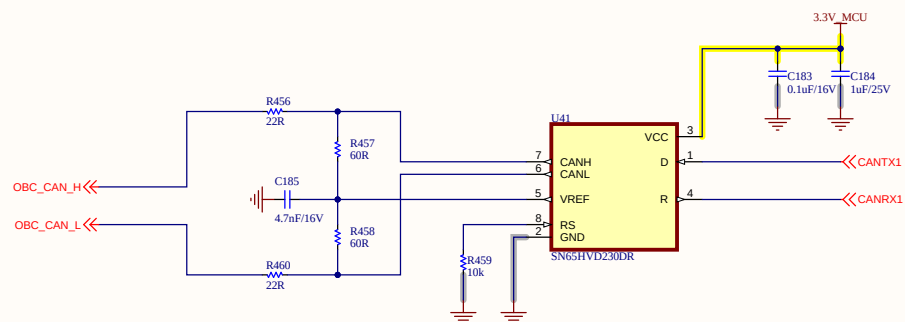
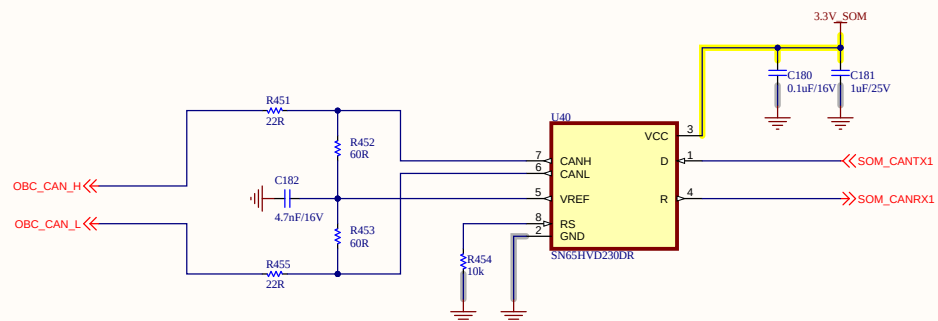
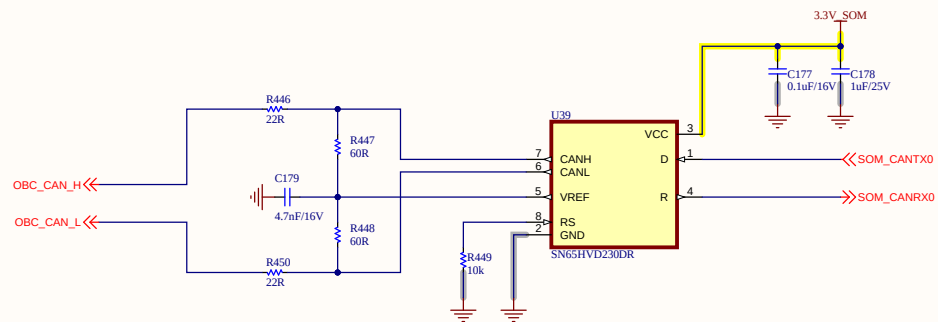
EFUSE for HD4



EFUSE for SLN

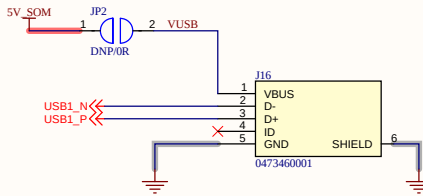


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 06_POWER_4.SchDoc	Sheet 10 of 25
Checked by: BAO BUI Q.	Document No: 10	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

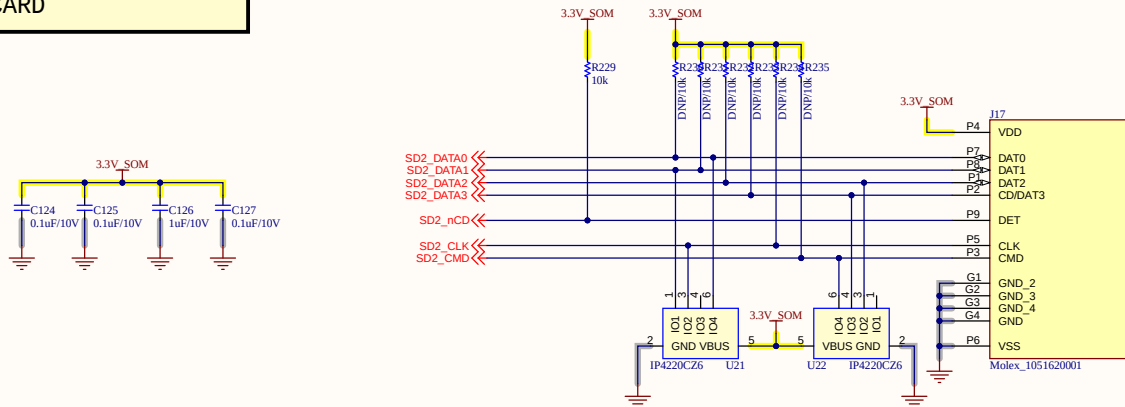


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 07_CAN.SchDoc	Sheet 11 of 25
Checked by: BAO BUI Q.	Document No: 11	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

USB

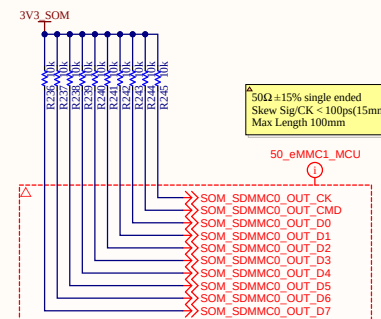
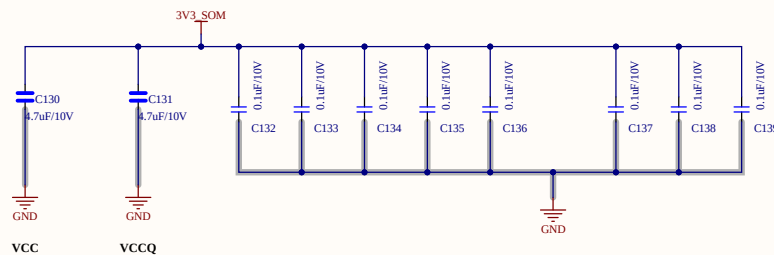
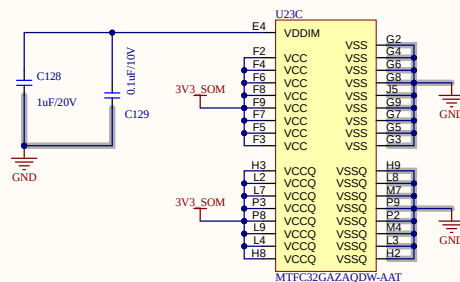
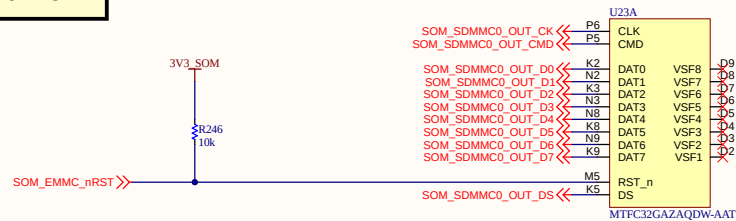


SD CARD

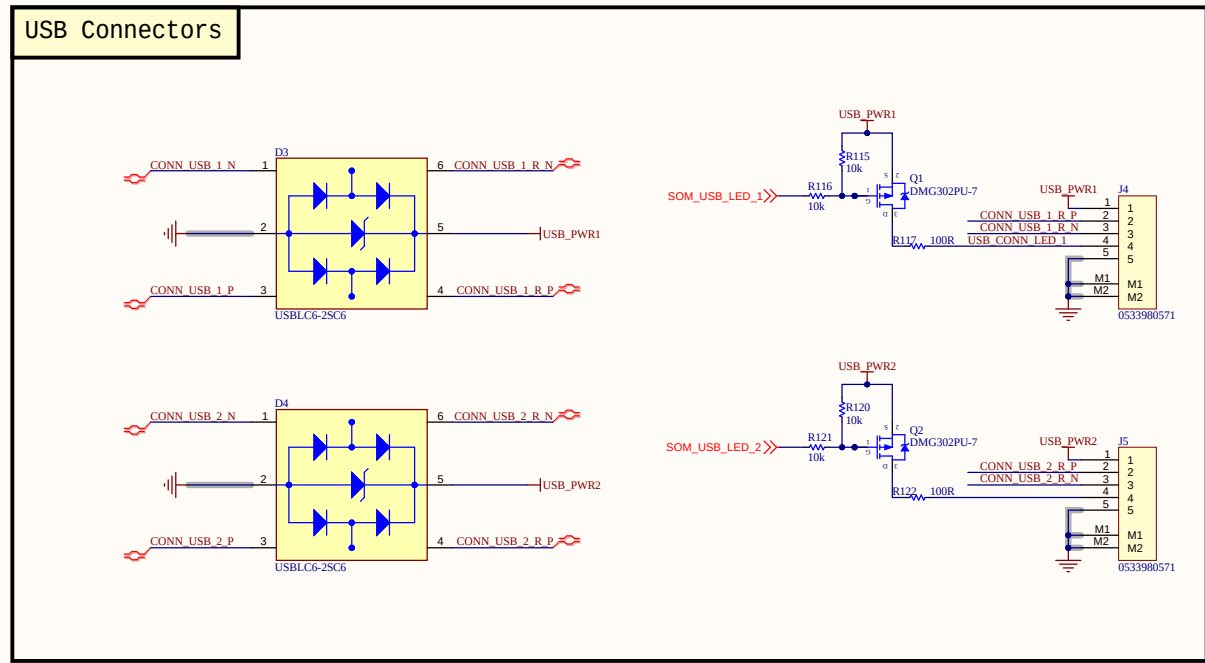
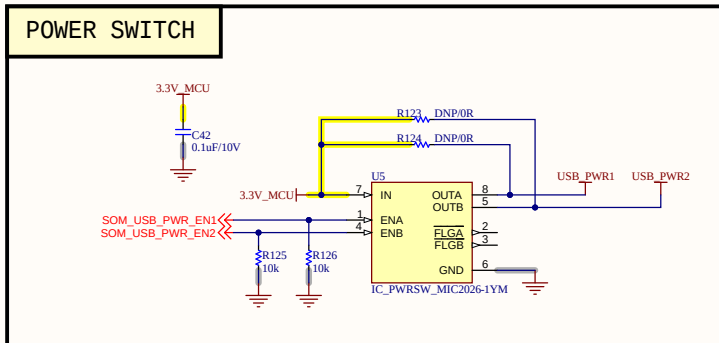
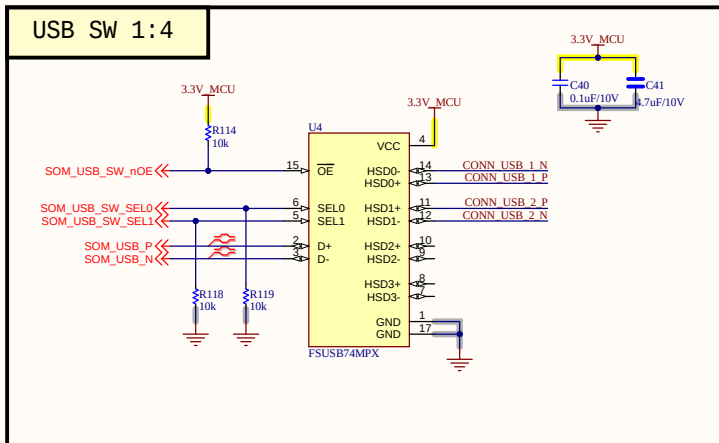


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 08_SOM_USB_CARD.SchDoc
Checked by: BAO BUI Q.	Document No: 13
Approved by: VU PHAM	Revision: V1.2.2

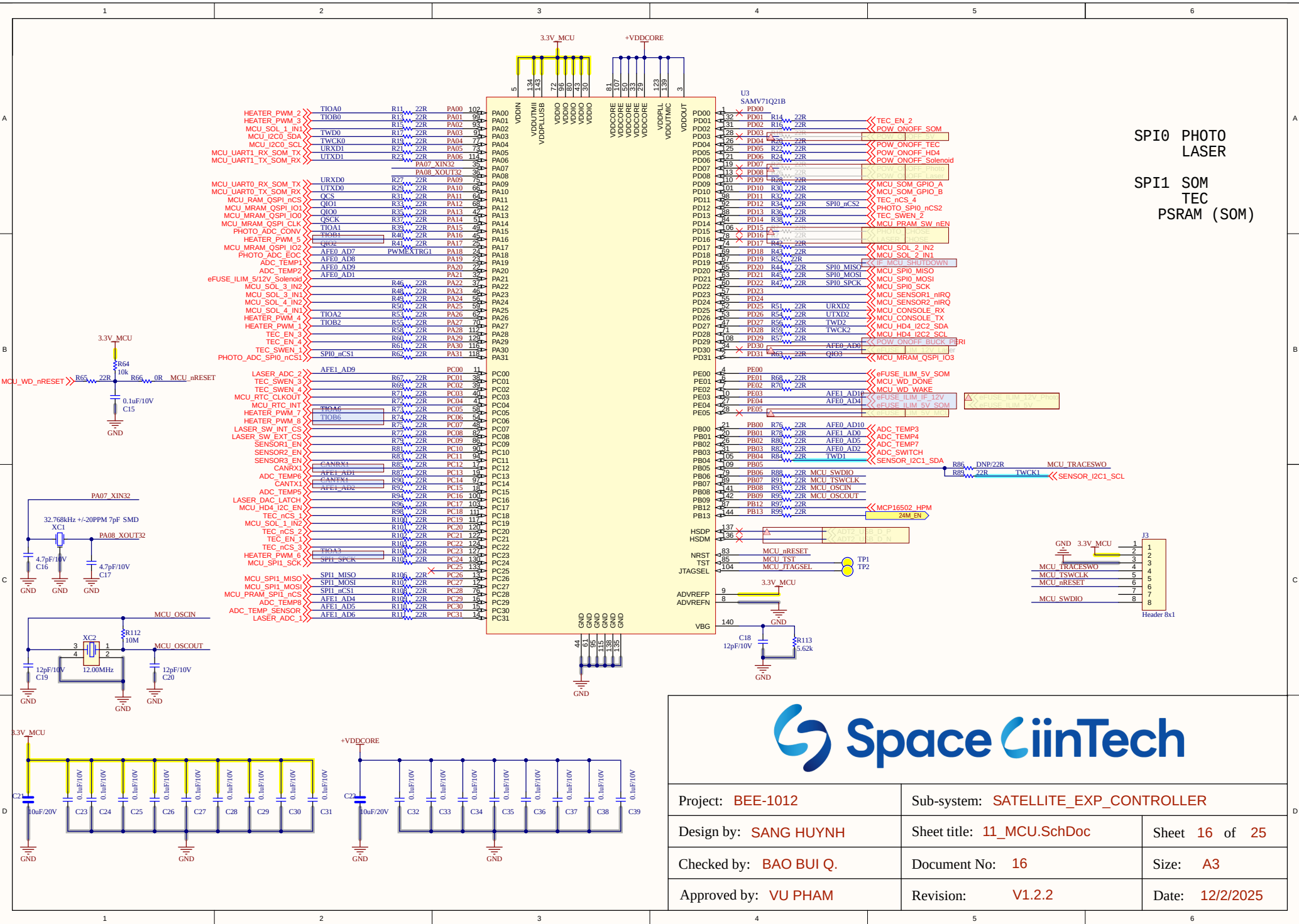
eMMC - Buffer



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 09_eMMC.SchDoc	Sheet 14 of 25
Checked by: BAO BUI Q.	Document No: 14	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025



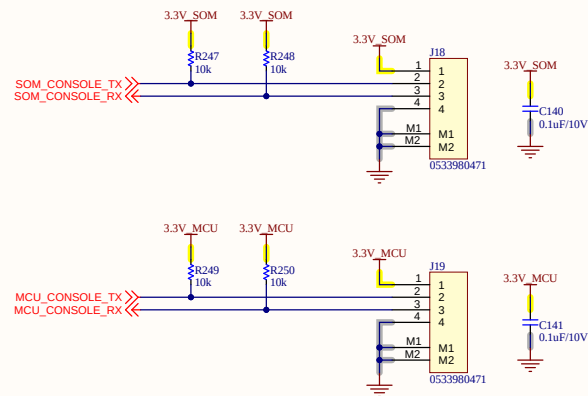
Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 10_USB.SchDoc	Sheet 15 of 25
Checked by: BAO BUI Q.	Document No: 15	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER		
Design by: SANG HUYNH	Sheet title: 11_MCU.SchDoc	Sheet 16 of 25	
Checked by: BAO BUI Q.	Document No: 16	Size: A3	
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025	

Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
--------	-----------	------------	------	------	------	------

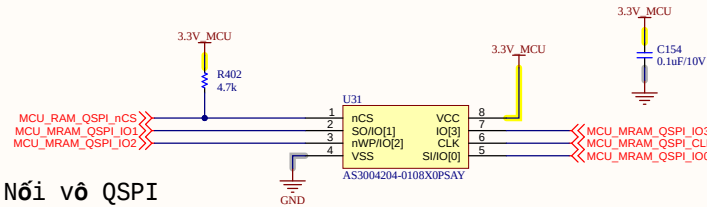
CONSOLE



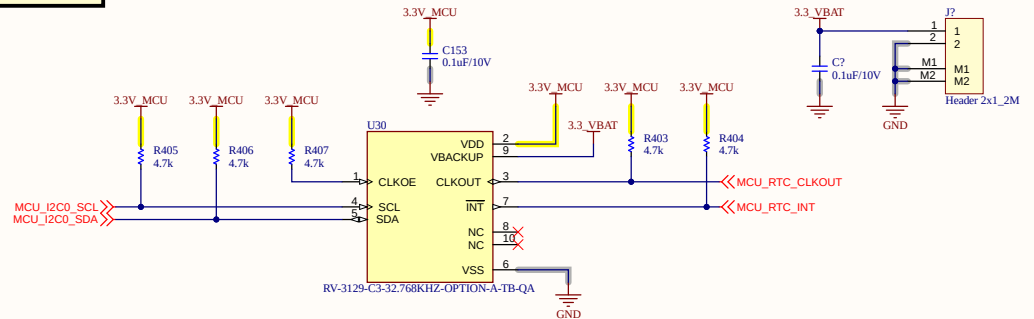
Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 12_CONSOLE.SchDoc	Sheet 17 of 25
Checked by: BAO BUI Q.	Document No: 17	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

MRAM 4Mbit

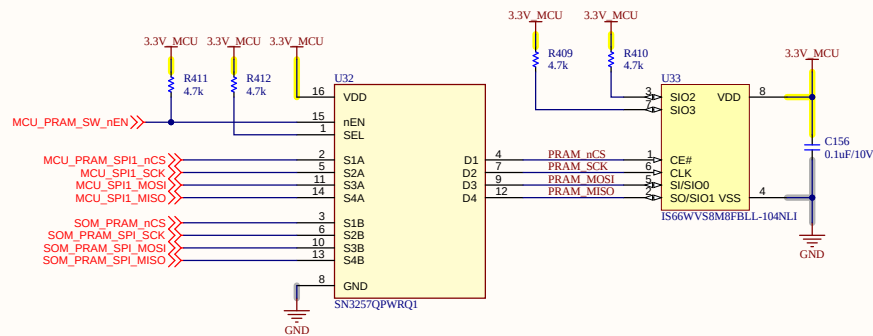
Nối vào QSPI



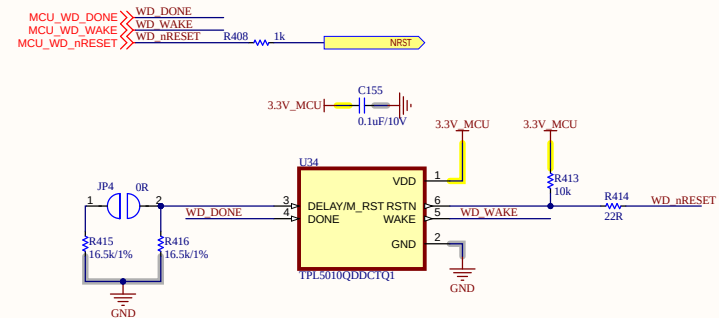
RTC



PRAM



WATCH DOG



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 13_MRAM_RTC_WDT.SchDoc Sheet 18 of 25

Checked by: BAO BUI Q.

Document No: 18

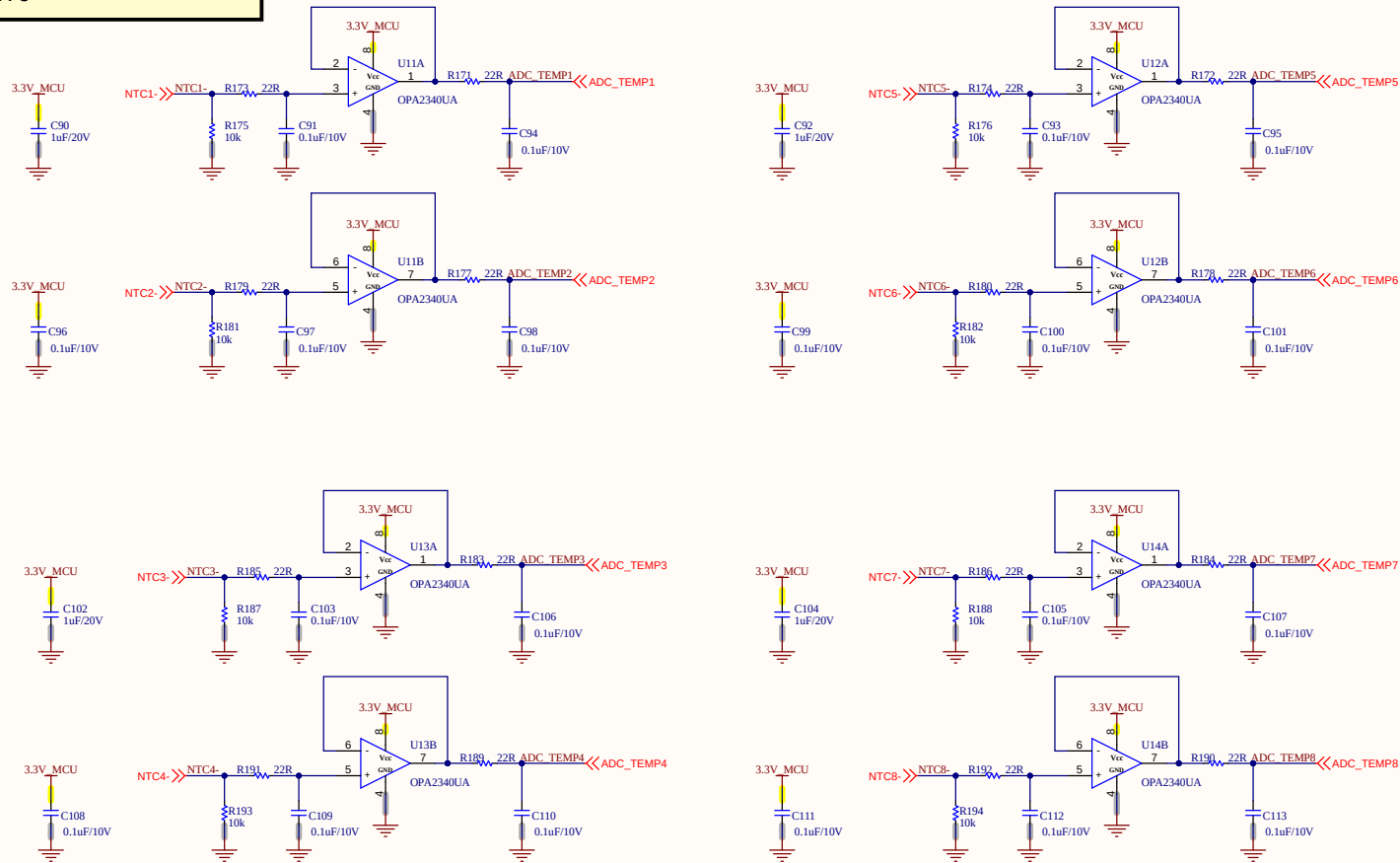
Size: A3

Approved by: VU PHAM

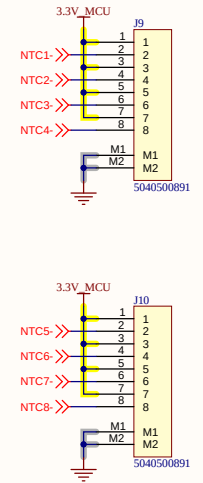
Revision: V1.2.2

Date: 12/2/2025

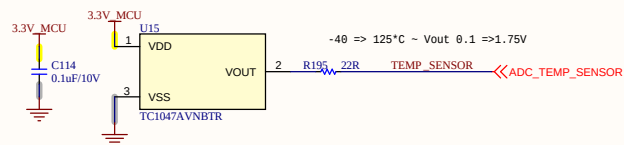
NTC



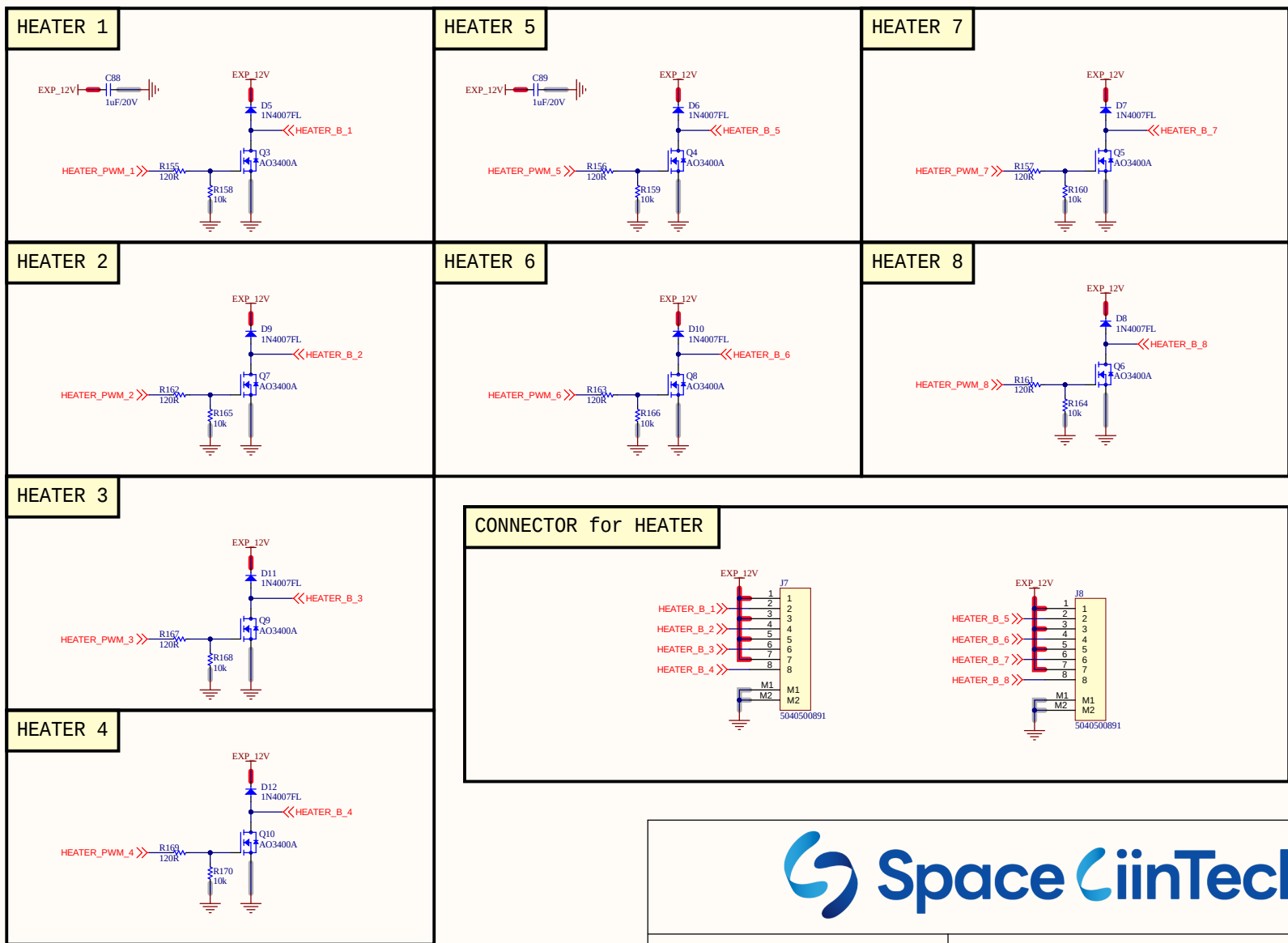
CONNECTOR FOR NTC



TEMPERATURE SENSOR

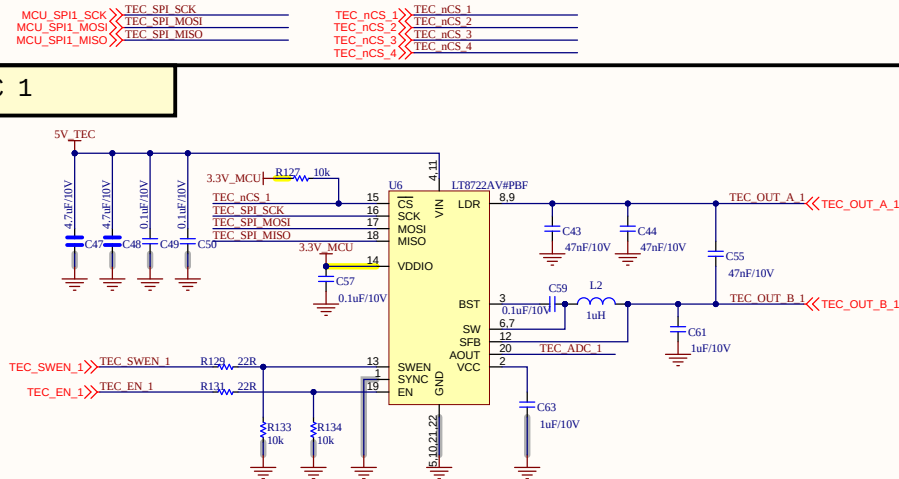


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 14_NTC_TEMPSENSOR_SCHDoc	19 of 25
Checked by: BAO BUI Q.	Document No: 19	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

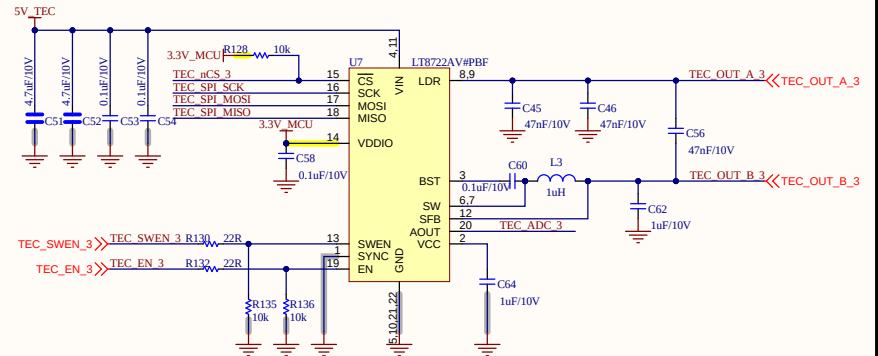


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 15_HEATER.SchDoc	Sheet 20 of 25
Checked by: BAO BUI Q.	Document No: 20	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

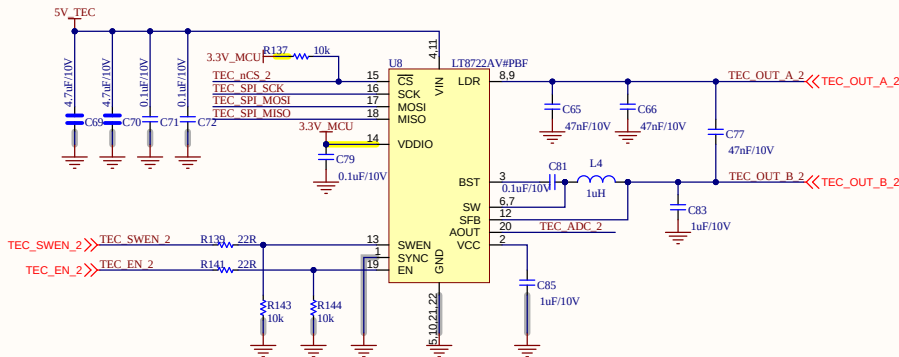
TEC 1



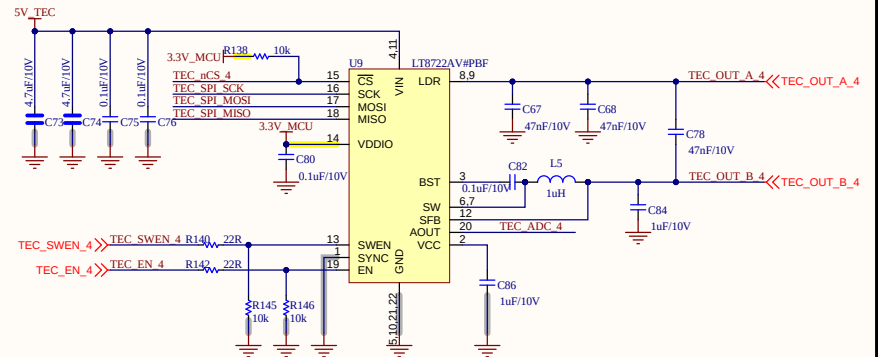
TEC 3



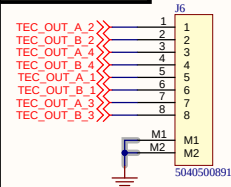
TEC 2



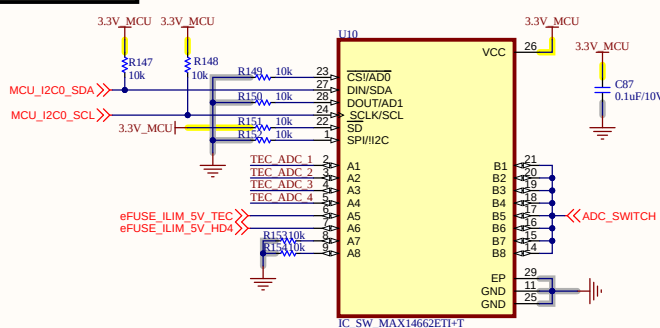
TEC 4



CONN TEC



CONN TEC



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 16_TEC.SchDoc

Sheet 21 of 25

Checked by: BAO BUI Q.

Document No: 21

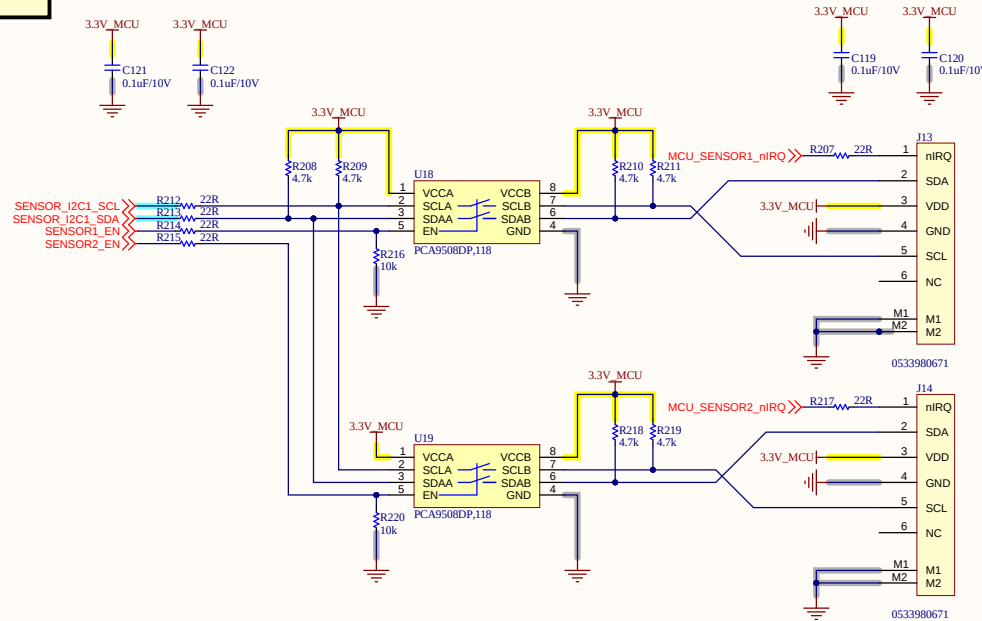
Size: A3

Approved by: VU PHAM

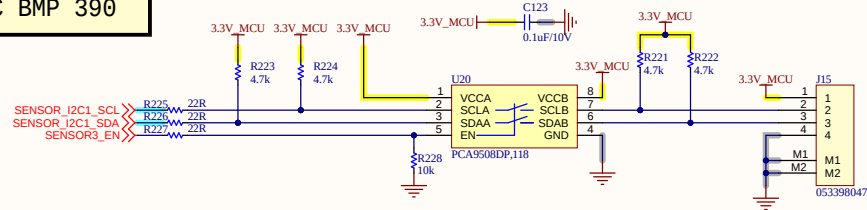
Revision: V1.2.2

Date: 12/2/2025

I2C

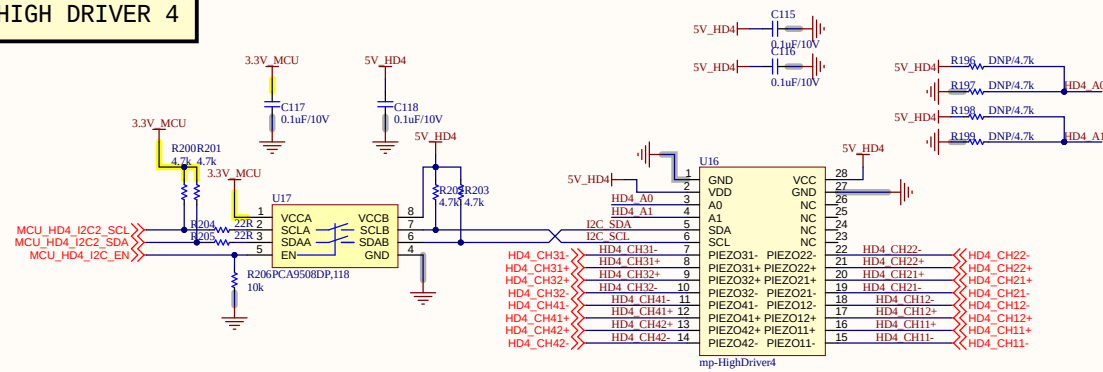


I2C BMP 390

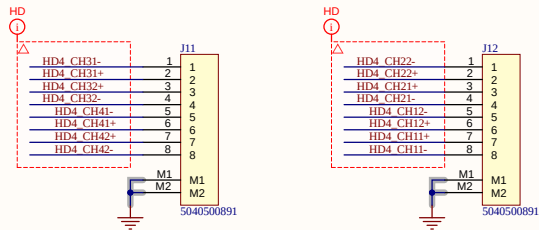


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 17_FLOW_Sensor.SchDocSheet 22 of 25	
Checked by: BAO BUI Q.	Document No: 22	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

HIGH DRIVER 4

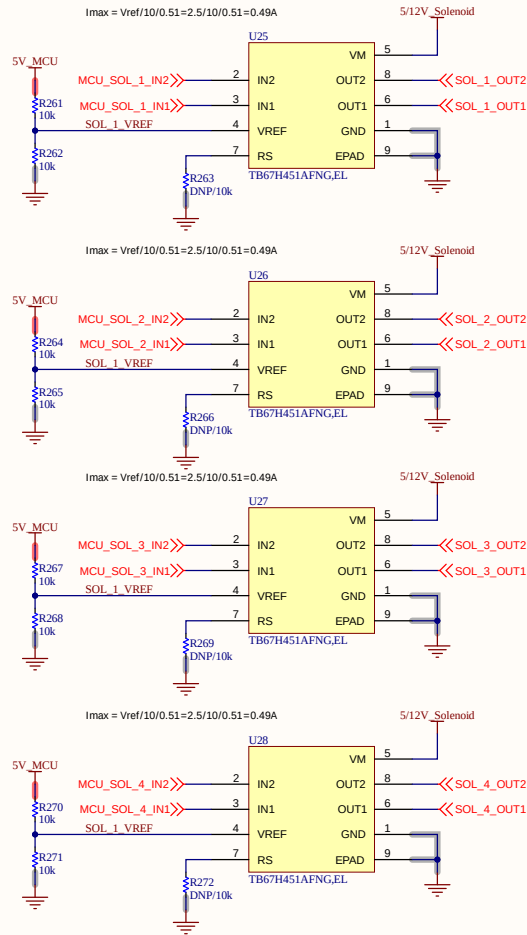


CONNECTOR BUMP

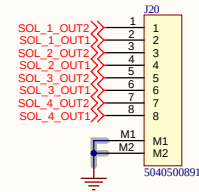


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 18_HD4_PUMP.SchDoc	Sheet 23 of 25
Checked by: BAO BUI Q.	Document No: 23	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

SOLENOID



CONNECTOR SOLENOID



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 19_SOLEDNOID.SchDoc Sheet 24 of 25

Checked by: BAO BUI Q.

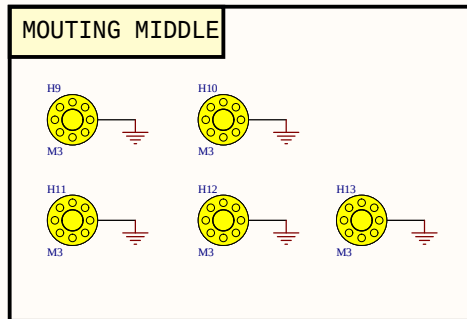
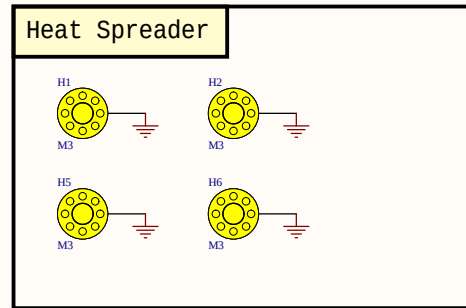
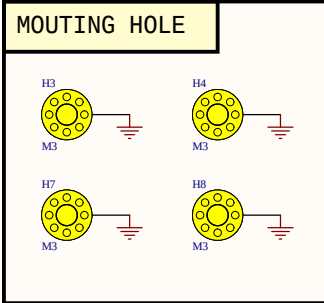
Document No: 24

Size: A3

Approved by: VU PHAM

Revision: V1.2.2

Date: 12/2/2025



Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: MOUTING.SchDoc	Sheet 25 of 25	
Checked by: BAO BUI Q.	Document No: 25	Size: A3	
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025	